Expectation, Variance & Indicator Variables — Complete Formula Reference

Every identity from Ch. 1.4, zero derivations, plus the extra results your 20-problem set actually used.

1. Linearity of Expectation

- **Core identity:** $\mathbb{E}\left[\sum_i a_i X_i\right] = \sum_i a_i \, \mathbb{E}[X_i]$
- **Holds under *arbitrary* dependence** — no independence, no knowledge of the joint distribution, no correlation structure required, ever
- **Affine shift:** $\mathbb{E}[aX + b] = a\,\mathbb{E}[X] + b$

trap: linearity is the only tool in this chapter that never needs independence. Every other formula below (variance, covariance) does care about dependence — don't let the ease of Section 1 lull you into forgetting that.

2. Variance & Covariance Algebra

- **Variance:** $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$
- **Covariance:** $\mathrm{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$
- **Variance of a sum (n vars):** $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i) + 2\sum_{i<j} \mathrm{Cov}(X_i, X_j)$
- **Two-variable combo:** $\mathrm{Var}(aX + bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X, Y)$
- **Independent case:** cross terms vanish, $\mathrm{Var}(\sum X_i) = \sum \mathrm{Var}(X_i)$
- **Equicorrelated portfolio (n vars, each Var$=\sigma^2$, pairwise Cov$=\rho\sigma^2$):**

$$\mathrm{Var}\left(\overline{X}\right) = \frac{\sigma^2[1 + (n-1)\rho]}{n} \xrightarrow{n\to\infty} \rho\sigma^2$$

trap: a negative covariance shrinks $Var(X+Y)$ but grows $Var(X-Y)$ by the same amount — a hedge only hedges the specific combination (long/long vs. long/short) it was built for. And diversification cannot drive portfolio variance to zero whenever $\rho > 0$ — it floors at $\rho\sigma^2$, no matter how large n gets.

3. The Fundamental Indicator Identity

- **Definition:** $\mathbb{I}_A \in \{0, 1\}, \quad \mathbb{E}[\mathbb{I}_A] = P(A)$
- **Idempotence:** $\mathbb{I}_A^2 = \mathbb{I}_A$ (squaring a 0/1 variable changes nothing)
- **Variance (follows free from idempotence):** $\mathrm{Var}(\mathbb{I}_A) = \mathbb{E}[\mathbb{I}_A] - (\mathbb{E}[\mathbb{I}_A])^2 = p(1-p)$
- **Covariance of two indicators:** $\mathrm{Cov}(\mathbb{I}_A, \mathbb{I}_B) = P(A \cap B) - P(A)P(B)$
- **Pair-indexed indicators:** for "how many *pairs* satisfy property X," index one indicator per pair ($\binom{n}{2}$ of them) rather than per item

trap: $\mathbb{I}_A^2 = \mathbb{I}_A$ is why indicator variance never needs a real second-moment calculation — $\mathbb{E}[\mathbb{I}_A^2]$ is always just $\mathbb{E}[\mathbb{I}_A]$ again. Forgetting this and trying to compute $\mathbb{E}[\mathbb{I}_A^2]$ "from scratch" wastes time for nothing.

4. Fixed Points & Cycles in Random Permutations

- **Expected # fixed points:** $\mathbb{E}[X] = 1$ for *every* $n \geq 1$ (one indicator per position, $P = \frac{1}{n}$ each, $n \cdot \frac{1}{n} = 1$)
- **Variance of # fixed points:** $\mathrm{Var}(X) = 1$ for *every* $n \geq 2$
- **General k-cycles:** $\mathbb{E}[\#\text{ of } k\text{-cycles}] = \frac{1}{k}$ for any fixed $k \leq n$ (fixed points are the k=1 case)
- **Connection:** the full distribution of # fixed points is governed by the derangement numbers D_n (Notebook 1) — but mean and variance ignore that complexity entirely

trap: mean = 1 and variance = 1 regardless of n is a genuinely surprising, parameter-free result — don't try to "sanity check" it away as wrong just because it looks suspiciously clean for a problem involving $n!$ permutations.

5. Runs & Alternating Patterns (Coin Flips)

- **Total runs, fair coin (n flips):** $\mathbb{E}[R] = 1 + (n-1) \cdot \frac{1}{2} = \frac{n+1}{2}$
- **Total runs, biased coin ($P(H) = p$):** $\mathbb{E}[R] = 1 + (n-1) \cdot 2p(1-p)$
- **Runs are maximized exactly at $p = \frac{1}{2}$:** $2p(1-p)$ peaks at $p = \frac{1}{2}$
- **Head-runs only, fair coin:** $\mathbb{E}[H] = \frac{1}{2} + (n-1) \cdot \frac{1}{4}$
- **Consistency check:** $\mathbb{E}$[head-runs] + $\mathbb{E}$[tail-runs] = $\mathbb{E}$[total runs] (always verify sub-counts sum to the whole)

trap: a highly biased coin produces fewer runs (long predictable streaks), not more — "runs" measure alternation/transitions, and transitions are rarest exactly when one outcome dominates.

6. Inversions in Random Permutations

- **Expected # inversions:** $\mathbb{E}[X] = \binom{n}{2} \cdot \frac{1}{2} = \frac{n(n-1)}{4}$ — always exactly half the maximum possible $\binom{n}{2}$
- **Variance:** $\mathrm{Var}(X) = \binom{n}{2} \cdot \frac{1}{4} + \binom{n}{3} \cdot \frac{1}{6} = \frac{n(n-1)(2n+5)}{72}$
- **Covariance structure (3 distinct indices $i<j<k$):** shared index plays the *same* role (both "smaller" or both "larger") $\Rightarrow$ Cov $= +\frac{1}{12}$; shared index is the *middle* of a chain $i<j<k$ $\Rightarrow$ Cov $= -\frac{1}{12}$; no shared index $\Rightarrow$ Cov $= 0$

trap: not every pair of overlapping indicators has the same covariance sign — you must classify which of the two overlap patterns you have before summing. This is the single most intricate covariance-bookkeeping result in the chapter; verify any new instance on small n by brute force before trusting it.

7. Coupon Collector's Problem

- **Decomposition:** $T = G_1 + \cdots + G_n$, $G_i \sim \mathrm{Geom}(p_i)$, $p_i = \frac{n-i+1}{n}$, **independent** stages
- **Mean:** $\mathbb{E}[T] = nH_n = n\sum_{j=1}^{n} \frac{1}{j} \approx n(\ln n + \gamma)$, $\gamma \approx 0.5772$
- **Variance (independent stages $\Rightarrow$ variances just add, no covariance bookkeeping):**

$$\mathrm{Var}(T) = \sum_{i=1}^{n} \mathrm{Var}(G_i) = \sum_{j=1}^{n} \frac{n(n-j)}{j^2}$$

$$\sim n^2 \cdot \frac{\pi^2}{6} \quad (\text{large } n)$$

trap: the last coupon dominates both the mean and the variance ($p_n = \frac{1}{n}$, expected n draws just for the final type) — the harmonic sum is "cheap start, catastrophic finish," not a uniform accumulation. Contrast with Section 6: because these stages are genuinely independent (not just identically-distributed-but-dependent like the inversion indicators), variance bookkeeping here is trivial addition, not covariance classification.

Quick Decision Guide — "Which Tool Do I Need?"

- Need $\mathbb{E}$[total] of several possibly-dependent quantities → **linearity**, always — never check independence first, it's irrelevant.
- Need Var or Cov of a sum → you *do* now need the dependence structure: get every pairwise $\mathrm{Cov}(X_i, X_j)$ (zero if independent).
- A random variable counts "how many items/pairs satisfy property P" → **indicator decomposition**: one $\mathbb{I}$ per item for per-item properties, one $\mathbb{I}$ per pair ($\binom{n}{2}$ of them) for symmetric pairwise properties.
- Computing Var of an indicator sum → use $\mathbb{I}^2 = \mathbb{I}$ to skip the second-moment step; the only real work is $\mathrm{Cov} = P(A \cap B) - P(A)P(B)$ for every overlapping pair.
- Many indicators, and they're **independent** (e.g. coupon-collector stages) → variances just add, skip covariance entirely.
- Many indicators, and they're **dependent but identically structured** (e.g. permutation fixed points, inversions) → classify every distinct *type* of overlap (shared index, shared role, chain vs. non-chain) and compute one representative covariance per type before summing.
- A permutation question asks about fixed points, k-cycles, or inversions and the answer looks suspiciously *independent of* n or exactly *half* of some maximum → that's not a coincidence, it's the signature of this chapter's elegant indicator results — trust it, but verify on small n if unsure.
- A coin-flip question asks about runs/streaks → convert to "does position i differ from position $i-1$," sum those local transition-indicators, add 1.

Numbers Worth Memorizing Cold

- Fixed points (any n): $\mathbb{E} = 1$, $\mathrm{Var} = 1$
- k-cycles (any $n \geq k$): $\mathbb{E} = 1/k$
- Inversions: $\mathbb{E} = \dfrac{n(n-1)}{4}$ (exactly half of $\binom{n}{2}$); $\mathrm{Var} = \dfrac{n(n-1)(2n+5)}{72}$
- Runs, fair coin: $\mathbb{E} = \dfrac{n+1}{2}$; head-runs only: $\mathbb{E} = \dfrac{1}{2} + \dfrac{n-1}{4}$
- Coupon collector: $\mathbb{E}[T] = nH_n$; $\mathrm{Var}(T) \sim n^2\pi^2/6$
- $\gamma \approx 0.5772$ (Euler–Mascheroni); $\pi^2/6 \approx 1.6449$ (Basel constant)
- Indicator covariance sign rule (inversions): same-role share $= +\frac{1}{12}$, chain share $= -\frac{1}{12}$, disjoint $= 0$